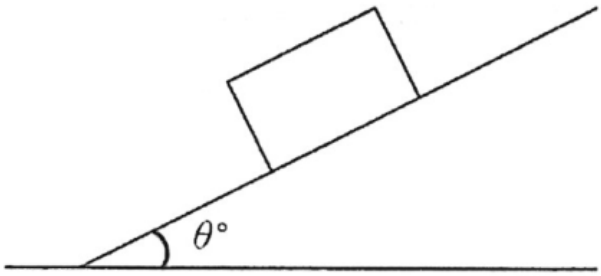
Forces: Inclined Planes

Questions 1–25

Question 1. The following question use $g = 9.8$ ms^{-2} A box of mass 5 kg rests on a rough slope which is inclined with an angle of $\theta°$.



The box is at the point of slipping.

a) Draw a force diagram.

Find the coefficient of friction between the box and the slope in each case when

b) $\theta = 30$

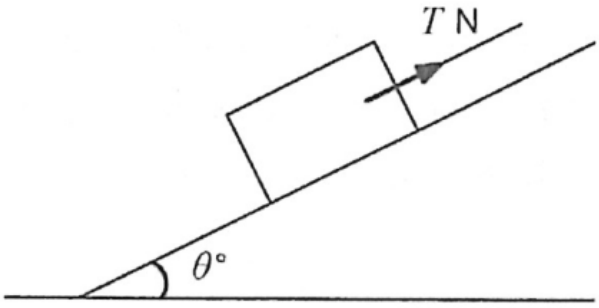
c) $\theta = 45$

d) $\theta = 55$

The box is replaced with a different box of mass m kg.

e) Show that $\mu = \tan\theta$

Question 2. A box of mass m kg rests on a rough slope which is inclined with an angle of $\theta°$. A pulling force T N acts upwards and parallel to the slope. A frictional force F N acts parallel to the slope.



The box is at the point of slipping up the slope.

a) Draw a force diagram.

Given

b) $m = 4, \mu = 0.8, \theta = 30$, find the value of F and the value of T

c) $m = 5, \mu = 0.6, \theta = 60$, find the value of F and the value of T

d) $T = 10, \theta = 20, F = 5$, find the value of m

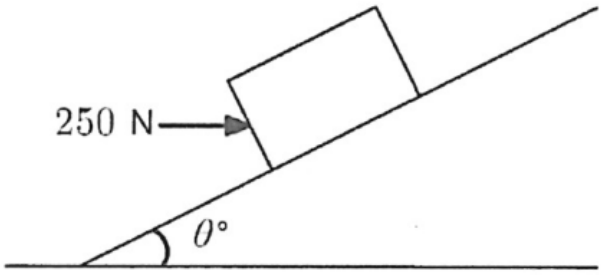
e) $T = 12, m = 2, F = 8$, find the value of θ

f) $T = 10, \mu = 0.6, \theta = 45$, find the value of m

Now using exact values of g

g) show that $T = mg(\mu\cos\theta + \sin\theta)$

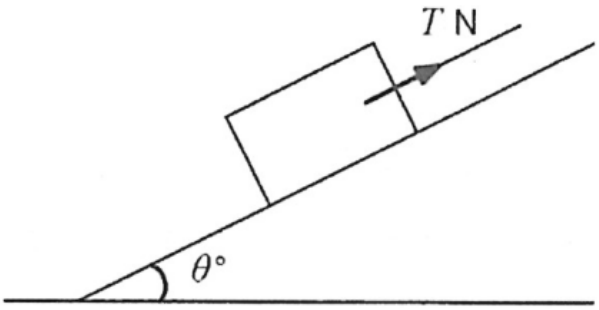
Question 3. A particle has mass 20 kg and is held at rest on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{3}{4}$



It is being held at rest by a horizontal pushing force 250 N.

a) Draw a force diagram.
b) Show that $\sin\theta = \frac{3}{5}$ and $\cos\theta = \frac{4}{5}$
c) Find the normal reaction force on the box.
d) Find the frictional force acting on the box.
e) Find the coefficient of friction acting between the box and the slope.

Question 4. A box of weight 100 N rests on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{7}{24}$



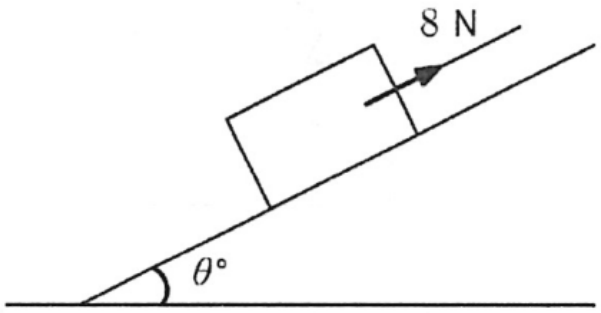
A pulling force T N acts upwards and parallel to the slope. The box is at the point of slipping down the slope.

a) Draw a force diagram.
b) Show that $\sin\theta = \frac{7}{25}$ and $\cos\theta = \frac{24}{25}$

The coefficient of friction between the box and the slope is 0.25

c) Find the value of T

Question 5. The following question use $g = 10\ \text{ms}^{-2}$ A box of mass 5 kg rests on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{2}{5}$



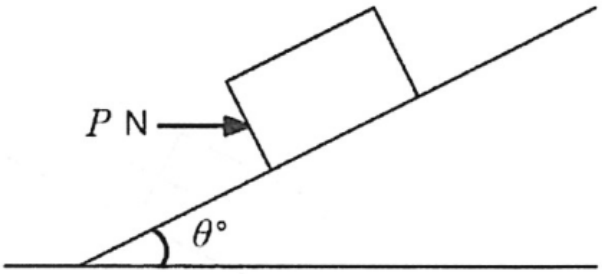
A pulling force 8 N acts upwards and parallel to the slope. The box is at the point of slipping down the slope.

a) Draw a force diagram.

b) Show that $\sin\theta = \frac{2}{\sqrt{29}}$ and $\cos\theta = \frac{5}{\sqrt{29}}$

c) Find the value of the coefficient of friction between the box and the slope.

Question 6. A particle has mass m kg and is held at rest on a rough slope which is inclined with an angle of $\theta°$.



It is being held at rest by a horizontal pushing force P N.

a) Draw a force diagram.

Given that $m = 2, \mu = 0.4, \theta = 30$, find

b) the normal reaction force on the box in terms of P

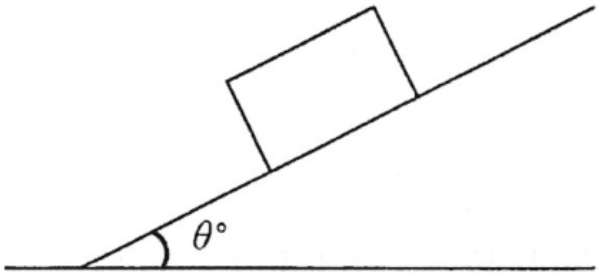
c) the frictional force in terms of P

d) the value of P

Question 7. A sledge is released from rest on a smooth slope which is at angle $10°$ to the horizontal. The mass of the sledge is m kg.

a) Draw a force diagram.
b) Find the acceleration of the sledge.
c) Find the speed once it has travelled 5 metres.

Question 8. A box of mass m kg is held at rest before being released. It lies on a rough slope which is inclined with an angle of $\theta°$. A frictional force F N acts parallel to the slope. The coefficient of friction between the box and the slope is μ



The acceleration of the box is given by a ms^{-2}.

a) Draw a force diagram.

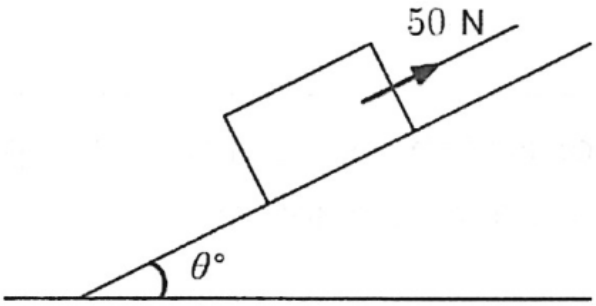
Given

b) $\theta = 60, m = 4, \mu = 0.2$, find the value of a and state its direction.
c) $\theta = 45, m = 12, a = 1.5$, find the value of μ

Now, using exact values of g,

d) show that the acceleration down the slope is given by $a = g(\sin\theta - \mu\cos\theta)$
e) State the acceleration down the slope if it were modelled as smooth.

Question 9. A box has weight 39 N, initially at rest and on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{5}{12}$. It is being pulled up the slope by a rope which may be modelled as a light, inextensible string which is parallel to the slope with the tension 50 N.



The coefficient of friction between the box and the slope is given by 0.5. The acceleration of the system is a ms^{-2}.

a) Draw a force diagram.

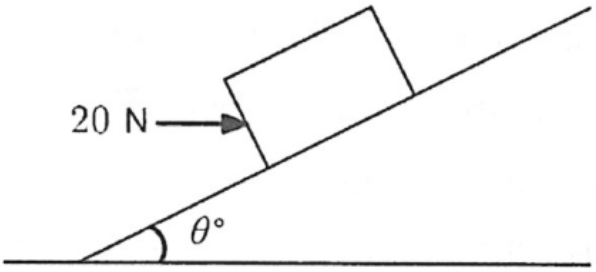
b) Show that $a = \frac{170}{39}$

The box is pulled up the slope for 1.5 seconds.

c) Find the distance travelled by the box in this time.

d) Find the speed it is moving at 1.5 seconds.

Question 10. The following questions use $g = 9.8$ ms^{-2} A particle has mass m kg and is held at rest on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{3}{7}$. It is being pushed up the slope by a horizontal force of 20 N which is horizontal.



The coefficient of friction between the box and the slope is given by μ. The acceleration of the system is a ms^{-2}.

a) Draw a force diagram.

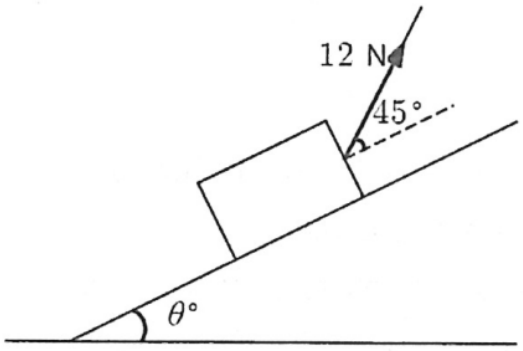
b) Find the exact values of $\cos\theta$ and $\sin\theta$

It is given that $m = 10$ and $\mu = 0.25$

c) Find the value of a

d) Describe the type and direction of the motion.

Question 11. A particle has mass m kg and is on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{7}{24}$
It is being pulled up the slope by a light, inextensible string which is at angle 45° to the slope with the tension 12 N.



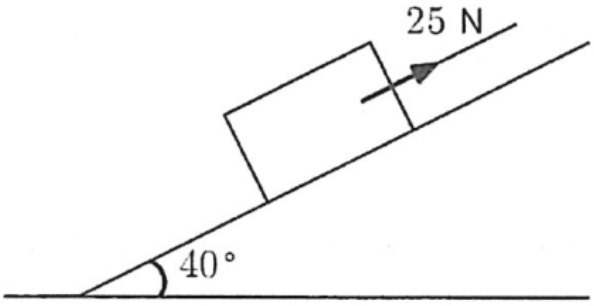
The coefficient of friction between the box and the slope is given by μ. The acceleration of the particle up the slope is 1.2 ms^{-2}.

a) Draw a force diagram.
b) Find the exact values of $\cos\theta$ and $\sin\theta$

Given

c) $m = 2$, find the value of μ
d) $\mu = 0.1$, find the value of m

Question 12. A crate has mass 6 kg on a rough slope which is inclined with an angle of 40°. It is being pulled up the slope by a rope with the tension 25 N which is parallel to the slope. The rope may be modelled as a light, inextensible string.



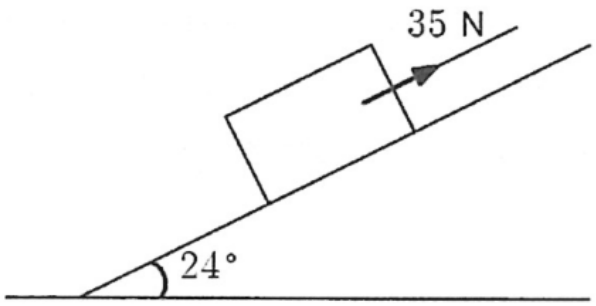
The coefficient of friction between the crate and the slope is 0.45.

a) Find the acceleration of the crate.
b) Describe the motion of the crate.

After coming to a stop, the rope is removed, and the crate slides down the slope.

c) Find the value of the acceleration down the slope.

Question 13. A crate has mass 4 kg, initially at rest and on a rough slope which is inclined with an angle of 24°. It is being pulled up the slope by a rope with the tension 35 N which is parallel to the slope. The rope may be modelled as a light, inextensible string.



The acceleration of the system is 0.5 ms^{-2}.

a) Show that the coefficient of friction between the box and the crate is approximately 0.48

b) Find how long it takes for the crate to move 2 metres up the slope.

Upon travelling 2 metres, the end of the rope which is not attached to the crate is instantaneously adjusted upwards so that it creates an angle of 20° with the slope but remains at 35 N.

c) Find how long it takes for the crate to move a further 2 metres up the slope.

Question 14. A toy car is placed at the end of a straight flat track of length 5 metres, held at an incline of $\theta°$ where $\tan\theta = \frac{3}{4}$. The toy car is released from rest and takes 2 seconds to reach the bottom of the track. The coefficient of friction between the car and the track is μ

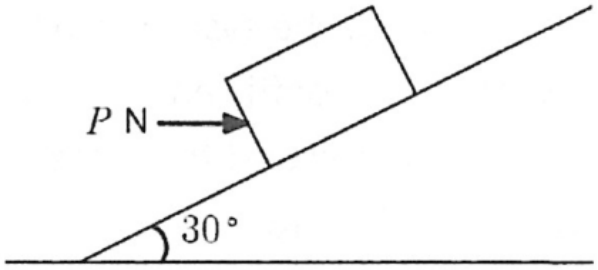
a) Find the acceleration of the toy car

b) Find the value of μ

Question 15. A particle of mass m kg is projected up a rough slope which has an incline of $\theta°$ where $\tan\theta = \frac{12}{5}$
The coefficient of friction between the particle and the slope is 0.8. The particle initially has speed 20 ms^{-1}.
Find the time it takes for the particle to return to its original position.

Question 16. A particle of mass m kg is projected up a rough slope which has an incline of $\theta°$ where $\tan\theta = \frac{3}{4}$.
It starts with speed 15 ms^{-1} and takes 3 s to return to its start position.
Find the coefficient of friction between the particle and the slope.

Question 17. A block has weight 20 N and is on a rough slope which is inclined with an angle of 30°. It is being pushed up the slope by a horizontal force of P N.



The normal reaction on the block by the slope is $15\sqrt{3}$ N. The block is decelerating up the slope at 2.45 ms^{-2}.

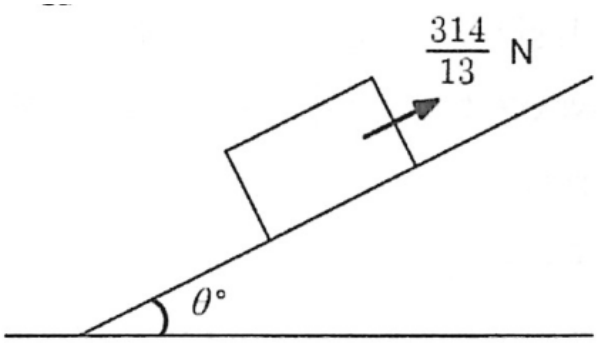
a) Find the friction force acting against the particle.

The horizontal force is removed.

b) Find the acceleration of the block.

c) Suggest improvements to the model in order to make the model more realistic.

Question 18. A particle has weight 49 N is on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{5}{12}$



The particle is pulled up the slope by a force parallel to the slope of $\frac{314}{13}$ N. The particle has deceleration 1.2 ms^{-2}.

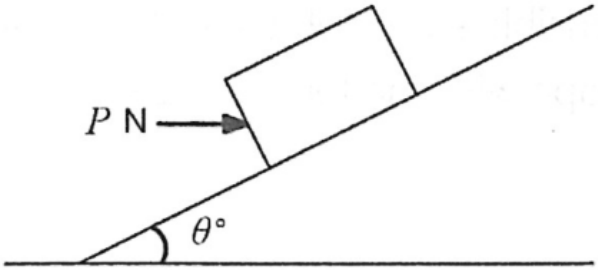
a) Find the coefficient of friction between the slope and the particle.

b) Given it takes 4 seconds to come to a stop, find the distance it travels up the slope.

The pulling force is then removed.

c) Find how long it takes for the particle to return to its original position from the point at which it came to a stop.

Question 19. The following questions use exact values of g A particle has mass m kg and is on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{4}{3}$. It is being pushed up the slope by a horizontal force of P N.



The coefficient of friction between the particle and the slope is 0.6

a) Find the values of P such that the particle is moving up the slope, giving your answer in terms of m and g

The particle is placed on a rough slope which is inclined at an angle of $\theta°$ where $\tan\theta = \frac{4}{3}$ with coefficient of friction μ

It is being pushed up the slope by a horizontal force of Q N.

b) Find the values of μ in terms of Q, m and g such that the particle will not accelerate up the slope.

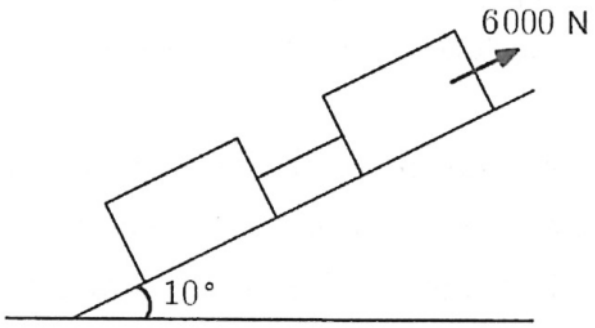
Question 20. Particle A of weight W N is on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{2}{3}$. The coefficient of friction between the particle and the slope is μ. The block is at the point of sliding down the slope.

a) Show that $\mu = \frac{2}{3}$

Particle A is replaced with particle B which has double the mass of A. The coefficient of friction remains the same.

b) Describe the motion of particle B.

Question 21. The following questions use $g = 10$ ms^{-2} A car of mass 2200 kg is towing a trailer of mass 400 kg along a straight flat road which is inclined at 10° to the horizontal. The driving force produced by the engine of the car is 6000 N.

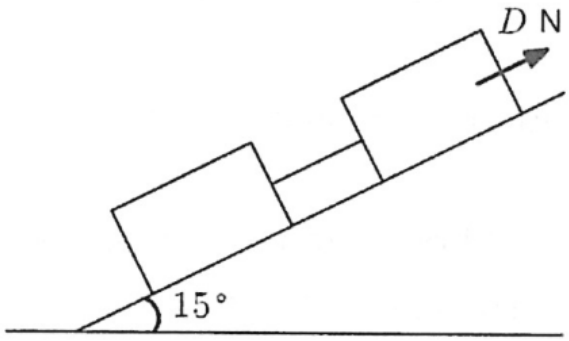


The coefficient of friction between the road and the vehicles is 0.1.

a) Find the deceleration of the car.

b) Find the tension in the tow bar.

Question 22. A van of mass 2400 kg is towing a trailer of mass 600 kg along a straight flat road which is inclined at 15° to the horizontal. The van is driving at 8 ms^{-1} and after 5 seconds has reached 12 ms^{-1} due to a driving force of D N.



The coefficient of friction between the road and the vehicles is 0.15.

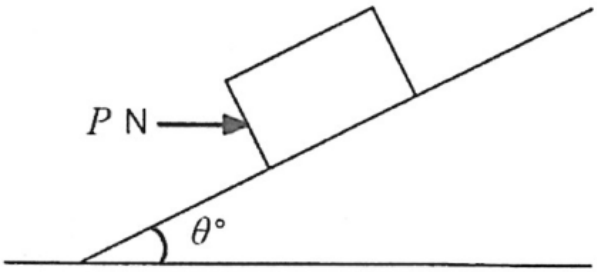
a) Find the value of D

b) Find the tension in the tow bar.

Question 23. The following questions use exact values of g A block A has mass m kg and is on a rough slope which is inclined with an angle of 45°. It is being pushed up the slope by a force of P N which is parallel to the slope.
The coefficient of friction between the block and the slope is μ. The block is at the point of sliding.
Show that $\mu = \frac{\sqrt{2}P}{mg} - 1$

Question 24. A particle of mass m kg lies on a rough slope which is inclined with an angle of $\theta°$.
It is being pulled upwards by a force of km N which is parallel to the slope, where k is a positive constant.
The coefficient of friction between the particle and the slope is μ
Show that $a = k - \mu g \cos\theta - g \sin\theta$

Question 25. A particle has weight w N and is on a rough slope which is inclined with an angle of $\theta°$ where $\tan\theta = \frac{7}{24}$. It is being pushed up the slope by a horizontal force of P N.
The coefficient of friction between the box and the slope is given by μ
The particle is at the point of sliding down the slope.



a) Draw a force diagram.
b) Show that $\mu = \frac{7w-24P}{24w+7P}$

The force P is removed and consequently the block moves down the slope with acceleration $\frac{1}{5}g$ ms^{-2}.
c) Show that $\mu = \frac{1}{12}$